

Auto-Load usb_storage Module at Boot

Objective: Ensure the `usb_storage` module is automatically loaded at every system boot, so it appears in `lsmod` even if no USB device is connected/attached.

below are the steps:

1. Verify the Module and ensure the module exists.

```
root@server1:~  
[root@server1 ~]# lsmod | grep usb_storage  
[root@server1 ~]# lsmod | grep usb  
[root@server1 ~]# modinfo usb_storage | grep -E 'filename|license|name|depends'  
filename:      /lib/modules/5.14.0-611.41.1.el9_7.x86_64/kernel/drivers/usb/storage/usb-storage.ko.xz  
license:       GPL  
depends:          
name:          usb_storage  
  
[root@server1 ~]#  
[root@server1 ~]#
```

2. make `usb_storage` load automatically at boot and it should be always show in `lsmod`.

modules can be loaded via `/etc/modules-load.d/`

```
root@server1:~  
[root@server1 ~]# modinfo usb_storage | grep -E 'filename|license|name|depends'  
filename:      /lib/modules/5.14.0-611.41.1.el9_7.x86_64/kernel/drivers/usb/storage/usb-storage.ko.xz  
license:       GPL  
depends:          
name:          usb_storage  
[root@server1 ~]# echo usb-storage | tee /etc/modules-load.d/usb-storage.conf  
usb-storage  
[root@server1 ~]#
```

In above step it writes the line '`usb_storage`' into the file `/etc/modules-load.d/usb-storage.conf`

we can also create file by `vi` n put above data into the `/usb_storage.conf` file

Explanation: `systemd-modules-load` reads this file at boot and ensures `usb_storage` is loaded automatically.

3. Verify Auto-loading- reboot the system

```
[root@server1 ~]#  
[root@server1 ~]# echo usb-storage | tee /etc/modules-load.d/usb-storage.conf  
usb-storage  
[root@server1 ~]# systemctl reboot
```

After reboot: check into lsmod

```
Activities Terminal Apr 8 2  
root@server1 ~]# lsmod | grep usb_storage  
usb_storage 94208 0  
root@server1 ~]# lsmod | grep usb  
usb_storage 94208 0  
root@server1 ~]#
```

This confirms the module is loaded, even if no USB device is connected.

Outcome: After following these steps, the usb_storage module will persistently load at every boot, appearing in lsmod even without a connected USB device.

note:

Module Blacklist: Ensure /etc/modprobe.d/ does not blacklist usb_storage

Kernel Upgrades: After kernel updates, confirm usb_storage is still available (modinfo usb_storage) and auto-load works.

Reference: man modules-load.d , man modprobe , chatgpt,man systemd-modules-load.service

#####

Some extra information:

Auto-Load Configuration by RHEL Version

rhel 6(init) : by /etc/rc.local

/etc/rc.local is deprecated in RHEL ecosystem-systemd replaces rc.local logic

rhel 8,9,10(systemd) : /etc/modules-load.d/

officially supported , used in enterprise automation (Ansible, Kickstart),

consistent across RHEL, Rocky, AlmaLinux

Use of /etc/modprobe.d/ vs /etc/modules-load.d/ in RHEL:

- **/etc/modules-load.d/ :**

purpose: Load modules automatically at boot

Managed by systemd-modules-load.service.

read all .conf file

Use this when you want a module always present, regardless of hardware events.

- **/etc/modprobe.d/ (Module Configuration)**

Purpose: use to Pass parameters/options to modules,Blacklist modules,Create aliases.

It does NOT automatically load modules at boot

Configure module behavior (options, blacklist, aliases).

Use this when you need to control module behavior, not to ensure it loads

How both Work Together:

/etc/modules-load.d/ → Triggers loading

/etc/modprobe.d/ → Controls how the module loads

Best Practice (Industry Standard):

<u>Scenario</u>	<u>Recommended</u>
<u>Approach</u>	
i. Always load module	/etc/modules-load.d/
ii. Tune module behavior	/etc/modprobe.d/
iii. Disable module	/etc/modprobe.d/ (blacklist)

iv. Production automation

Manage both via config management tools